MATH 3603–Homework 4

Due: 25-April-2026

1. Recall our definition: A stochastic process $\{X_n, n = 0, 1, 2, \cdots\}$ with state space $S = \{0, 1, 2, \cdots, l\}$ is called a Markov chain if for any $i, j, i_0, i_1, \cdots, i_{n-1} \in S$ and $n \geq 0$,

$$P\big(X_{n+1} = j | X_n = i, X_{n-1} = i_{n-1}, \cdots, X_1 = i_1, X_0 = i_0\big) = P\big(X_1 = j | X_0 = i\big).$$

Then answer the following questions.
(a) Use this definition to prove that for the Markov chain defined as above, and for $n > k \geq 0$

$$P\big(X_n = j | X_k = i, X_{k-1} \neq i_{k-1}, \cdots, X_1 \neq i_1, X_0 \neq i_0\big) = P\big(X_{n-k} = j | X_0 = i\big).$$

(b) For the Markov chain defined as above and for $n > k \geq 0$, does

$$P\big(X_n = j | X_k \neq i, X_{k-1} = i_{k-1}, \cdots, X_1 = i_1, X_0 = i_0\big) = P\big(X_{n-k} = j | X_0 \neq i\big)?$$

Prove it or disprove it.

2. A Markov chain $\{X_n, n \geq 0\}$ with states 0,1,2, has transition probability matrix

$$\begin{bmatrix} \frac{1}{2} & \frac{1}{3} & \frac{1}{6} \\ 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{2}{3} & 0 & \frac{1}{3} \end{bmatrix}$$

If $\mathbb{P}(X_0 = 0) = \mathbb{P}(X_0 = 1) = \frac{1}{4}$, find $\mathbb{E}[X_3]$.

3. Suppose that coin 1 has probability 0.7 of coming up heads, and coin 2 has probability 0.6 of coming up heads. If the coin flipped today comes up heads, then we select coin 1 to flip tomorrow, and if it comes up tails, then we select coin 2 to flip tomorrow. If the coin initially flipped is equally likely to be coin 1 or coin 2, then what is the probability that the coin flipped on the third day after the initial flip is coin 1?

4. Specify the classes of the following Markov chains (which states are in the same class), and determine for all states whether they are transient or recurrent:

$$P_1 = \begin{bmatrix} 0 & \frac{1}{3} & \frac{2}{3} \\ \frac{1}{2} & 0 & \frac{1}{2} \\ \frac{1}{4} & \frac{3}{4} & 0 \end{bmatrix}, \qquad P_2 = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 \\ \frac{1}{3} & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 1 & 0 \end{bmatrix}$$

$$P_3 = \begin{bmatrix} \frac{1}{2} & 0 & \frac{1}{2} & 0 & 0 \\ \frac{1}{4} & \frac{1}{2} & \frac{1}{4} & 0 & 0 \\ \frac{1}{3} & 0 & \frac{2}{3} & 0 & 0 \\ 0 & 0 & 0 & \frac{2}{3} & \frac{1}{3} \\ 0 & 0 & 0 & \frac{1}{2} & \frac{1}{2} \end{bmatrix}$$

5. Let $\xi_1, \xi_2, \ldots,$ be i.i.d. random variables, taking values from $\{1, 2, \cdots, N\}$ and taking each value with probability $1/N$. Show that $X_n = |\{\xi_1, \ldots, \xi_n\}|$ (cardinality of the set) is a Markov Chain and compute its transition probability.

6.: Show that in the Ehrenfest chain, all states are recurrent.